Algebra 1: Unit 9, Lesson 6

1. Sofia solved the equation, $x^2 - 8x - 43 = 0$, by completing the square. Some of her work is shown. Fill the blanks with the steps that complete Sofia's work.

   $x^2 - 8x - 43 = 0$

   $(x^2 - 8x) - 43 = 0$

   $(x^2 - 8x + 16) - 43 - 16 = 0$

   $(x - \underline{\hspace{2cm}})^2 - 43 - 16 = 0$

2. Complete the missing step in the table to show the next step when solving by completing the square.

Original Equation	$x^2 - 24x + 14 = 0$
Step 1	$(x^2 - 24x + \underline{\hspace{2cm}}) + 14 + \underline{\hspace{2cm}} = 0$

3. Fill the blank with the step that completes the missing step to complete the square.

   $x^2 - 10x - 38 = 0$

   $(x^2 - 10x) - 38 = 0$

   $(x^2 - 10x + \underline{\hspace{2cm}}) - 38 - 25 = 0$

   $(x - 5)^2 - 63 = 0$